



Low-concentration exposure to glyphosate-based herbicide modulates the complexes of the mitochondrial respiratory chain and induces mitochondrial hyperpolarization in the *Danio rerio* brain

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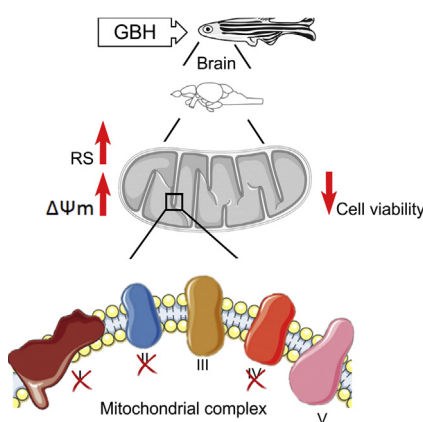
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HIGHLIGHTS

- LC_{50,96h} for adult zebrafish *Danio rerio* was 53.75 (± 3.30) mg L⁻¹ fGLY.
- *In vivo* low-concentrations 0.065, 1.0 mg L⁻¹ fGLY induced cell viability reduction.
- Inhibition of complex I and IV activity was detected at 0.065 and 1.0 mg L⁻¹ fGLY.
- Mitochondrial hyperpolarization was observed at 0.065; 1.0 and 10.0 mg L⁻¹ fGLY.
- Behavioral impairments were observed at low concentration (0.065 mg L⁻¹ fGLY).

GRAPHICAL ABSTRACT



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ABSTRACT

Glyphosate (N-phosphonomethyl-glycine) (GLY) is the active ingredient of the most used herbicides in the world. GLY is applied in formulated products known as glyphosate-based herbicides (GBH), which could induce effects that are not predicted by toxicity assays with pure GLY. This herbicide is classified as organophosphorus compound, which is known to induce neurotoxic effects. Although this compound is classified as non-neurotoxic by regulatory agencies, acute exposure to GBH causes neurological symptoms in humans. However, there is no consensus in relation to neurotoxic effects of GBH. Thus, the aim of this study was to investigate the neurotoxic effects of the GBH in the zebrafish *Danio rerio*, focusing on acute toxicity, the activity and transcript levels of mitochondrial respiratory chain complexes, mitochondrial membrane potential, reactive species (RS) formation, and behavioral repertoire. Adult zebrafish were exposed *in vivo* to three concentrations of GBH Scout[®], which contained GLY in formulation (fGLY) (0.065, 1.0 and 10.0 mg L⁻¹ fGLY) for 7 d, and an *in vitro* assay was performed using also pure GLY. Our results show that GBH induced in zebrafish brain a decrease in cell viability, inhibited mitochondrial

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Transcript level
Behavioral
Zebrafish

complex enzymatic activity, modulated gene expression related to mitochondrial complexes, induced an increase in RS production, promoted hyperpolarization of mitochondrial membrane, and induced behavioral impairments. Together, our data contributes to the knowledge of the neurotoxic effects of GBH. Mitochondrial dysfunction has been recognized as a relevant cellular response that should not be disregarded. Moreover, this study pointed to the mitochondria as an important target of GBH.

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1. Introduction

Pesticides are used extensively, and glyphosate (N-phosphonomethyl-glycine) (GLY) is the active ingredient of the most widely used herbicides in the world. The action of GLY results from its inhibition of a key plant enzyme, 5-enolpyruvylshikimate-3-phosphate synthase, which is involved in the synthesis of aromatic amino acids (Rueppel et al., 1977; Sharps, 1984). Since this enzyme is not present in animals, it has long been believed that GLY does not affect non-target animals (Williams et al., 2000). However, considerable evidence suggests that this argument could be imprecise, mainly because GLY is applied in formulated products, and such formulations could cause effects not predicted by toxicity assays using pure GLY (Vandenberg et al., 2017).

Since the commercial application of glyphosate-based herbicides (GBH) began four decades ago, their use has considerably expanded and diversified (Vandenberg et al., 2017). The global use of GLY has risen almost 15-fold after genetically modified GLY-tolerant crops were introduced in 1996 (Benbrook, 2016), added to the need for successive applications after the appearance of the GLY-resistant weeds, as well as the increased use as a desiccant in the pre-harvest of different crops (Guyton et al., 2015; Townsend et al., 2017). Thus, the intensive use of GLY has increased its residue levels in products destined for food, and in water (Hanke et al., 2010; Böhn et al., 2014). In human studies, GLY and its main metabolite aminomethylphosphonic acid (AMPA) have been detected in the blood, urine, and in maternal and umbilical cord blood (Acquavella et al., 2004; Conrad et al., 2017; Kongtip et al., 2017). Moreover, accidental intoxication by GBH leads to neurological symptoms such as shaking, loss of coordination, tingling, neuropathy, and ataxia (Barbosa et al., 2001; Hawkins and Cordova, 2009; Wang et al., 2011).

Although GBH use has increased dramatically, most of the research used in the risk assessment process to support its safety was conducted more than 30 years ago (US EPA, 1994), and current safety standards for GBH are outdated and may fail to protect public health or the environment (Vandenberg et al., 2017).

GLY is classified as an organophosphorus compound, and these compounds have been found to induce neurotoxic effects (Burns et al., 2013). However, there is a controversy regarding these neurotoxic effects, since international agencies classify GLY as a non-neurotoxic compound, even at very high doses (US EPA, 1994), and claim that it is unlikely to be neurotoxic (EFSA, 2015), which conflicts with a number of scientific studies (Coullery et al., 2016; Roy et al., 2016; Martínez et al., 2018).

Regarding GBH, the effects of these compounds have been described on acetylcholinesterase activity in brain, commonly used as a neurotoxicity biomarker (Menéndez-Helman et al., 2012; Braz-Mota et al., 2015). The toxicological profile of GBH has been investigated using *in vivo* and *in vitro* assays with different organisms, such as *Caenorhabditis elegans*, rodents, and fish. The major impairments are related to cytotoxicity of the cholinergic, glutamatergic, dopaminergic, and GABAergic neurons, in exposed adults and/or maternally exposed offspring (Negga et al., 2011; Cattani

et al., 2014, 2017).

Neurons seemed to be the main target in the studies that evaluated GLY and GBH neurotoxicity. However, it is known that mitochondria are impaired by many agrochemicals (Boelsterli, 2007; Mostafalou and Abdollahi, 2013). The main mitochondrial alterations studied are related to increased reactive oxygen species (ROS) production, swelling, and anti-oxidant system inhibition in different organisms and tissues (Luo et al., 2017; Bailey et al., 2018). The function of the central nervous system (CNS) strongly depends on efficient mitochondrial function, because brain tissue has a high energy demand (Federico et al., 2012); thus, the use of mitochondrial parameters is an interesting approach to better understand the neurotoxicity of GBH. Considering the extensive use of GBH, which has led to environmental and food contamination for non-target organisms, and the controversy in relation to its neurotoxic effects, we propose to investigate the neurotoxic effects of the GBH in brain cells of zebrafish *Danio rerio*, focusing on cell viability, enzymatic activity and transcript levels of mitochondrial complexes, reactive species formation, and mitochondrial membrane potential, as well as on fish behavior.

2. Materials and methods

2.1. Herbicide

The commercial formulation Scout® (Monsanto do Brasil Ltda), a GBH that contains 720 g kg⁻¹ GLY was used. The GBH dilutions were prepared in dechlorinated water, considering the amount of GLY contained in Scout® formulation, which are referred as fGLY (mg L⁻¹) to perform *in vivo* and *in vitro* assays. Pure GLY [N-phosphonomethyl-glycine 96%, Sigma-Aldrich] was prepared in dechlorinated water for *in vitro* assays, which is referred as GLY (mg L⁻¹).

2.2. Fish maintenance

Wild type adult zebrafish (*D. rerio*; body mass of 0.46 g ± 0.14 and length of 3.3 cm ± 0.09) were obtained from an aquarium shop, and kept in a 20 L aquarium with dechlorinated and filtered water. The fish were acclimated for 7 d under continuous aeration, at a controlled temperature of 27 ± 1 °C, pH 7.0 (±0.2), at a storage density of 1.0 g fish L⁻¹, and exposed to a light: dark cycle of 14:10 h, according to Lawrence (2007). The fish were fed once a day with commercial fish food. All experiments were approved by the Ethical Committee for Animal Use from Universidade Federal de Santa Catarina; CEUA-UFSC protocol No. 5466040416/2016.

2.3. Determination of the median lethal concentration (LC₅₀)

Acute toxicity bioassays were carried out to determine the LC_{50,96h} value of GBH for zebrafish. This test was performed according to the Organization for Economic Co-operation and Development (OECD) guideline test 203, under semi-static renewal conditions, with complete renewal of the test solution every 24 h.

Nominal concentrations of active ingredient tested were 0.0 (control), 0.1, 1.0, 10.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0, and 100.0 mg L⁻¹ fGLY. Each test was performed in triplicate, with eight fish randomly exposed to each concentration. Mortality was recorded after exposure at 24, 48, 72, and 96 h of exposure. During the assay, the dead fish were counted and removed from the aquarium. After 96 h of exposure, the LC₅₀ value was calculated using United States Environmental Protection Agency Probit software version 1.5 (US EPA, 1992).

2.4. GBH exposure in zebrafish

To evaluate the effects of GBH in the zebrafish brain, fish were randomly divided into four groups: three groups were exposed to GBH concentrations of 0.065 mg L⁻¹ fGLY (~0.12% LC_{50,96h}), 1.0 mg L⁻¹ fGLY (~2% LC_{50,96h}), and 10.0 mg L⁻¹ fGLY (~20% LC_{50,96h}), and the fourth group was exposed to 0.0 mg L⁻¹ fGLY (control group). The concentration of 0.065 mg L⁻¹ fGLY tested is the maximum concentration established for Brazilian inland waters (CONAMA, 2005; Resolution No. 357-Brazil). The concentrations of 1.0 and 10.0 mg L⁻¹ GLY were referenced in previous studies to GLY and/or GBH toxicity in fish (Modesto and Martinez, 2010; Uren Webster et al., 2014; Sulukan et al., 2017). The fish were exposed for 7 d. During the exposure period, the herbicide concentration was totally renewed in the aquarium 4 d after starting the experiment to maintain the expected GLY concentration.

2.5. Cell viability test

The MTT test (3-(4,5)-dimethylthiazolyl-2,5-diphenyltetrazolium bromide) (Mosmann, 1983) was used to evaluate brain cell viability. The *in vitro* assay was performed with GLY and fGLY after 3 h of incubation, and the *in vivo* assay was performed after 7 d of formulation exposure. For both assays, fish were killed by decapitation (CONCEA, 2013) and the brain was rapidly dissected (n = 6 animals per group), weighed and maintained in 1 mL ice cold Krebs-Ringer bicarbonate buffer (KRB) (122.0 mM NaCl, 3.0 mM KCl, 1.2 mM MgSO₄, 1.3 mM CaCl₂, 0.4 mM KH₂PO₄, 25.0 mM NaHCO₃, and 10.0 mM glucose). The buffer was bubbled with 95% O₂ and 5% CO₂ at pH 7.4. For the *in vitro* assay, the brains were incubated for 3 h in graded concentrations of GLY and fGLY [0.0 (control); 0.065 mg L⁻¹ (0.38 μM); 1.0 mg L⁻¹ (6.0 μM); 10.0 mg L⁻¹ (60.0 μM); 160.0 mg L⁻¹ (1.0 mM); 1.6 × 10³ mg L⁻¹ (10.0 mM); 4.0 × 10³ mg L⁻¹ (25.0 mM) and 8.0 × 10³ mg L⁻¹ (50.0 mM)] diluted in KRB buffer. Positive controls were used: 40.0 mg L⁻¹ rotenone (100.0 μM) (mitochondrial complex I inhibitor) and 3.0 × 10³ mg L⁻¹ sodium azide (50.0 mM) (mitochondrial complex II/IV inhibitor). After the incubation period, the samples were washed with KRB and incubated with MTT (1.2 mM, Sigma Aldrich) in 96-well plates for 20 min at 27 °C. For the *in vivo* assay, after the period of fish exposure to 0.065, 1.0 and 10.0 mg L⁻¹ fGLY, the brains were rapidly dissected, weighed and maintained in 1 mL ice cold KRB. MTT (1.2 mM, Sigma Aldrich) was added and the samples were incubated for 20 min at 27 °C. Then, samples from the *in vitro* and *in vivo* assays were washed in KRB, and 200.0 μL 1 M dimethyl sulfoxide (DMSO, Nuclear) was added, followed by 30 min of agitation. Then, 100.0 μL of formazan solubilized in DMSO was removed and added to each well of a new plate, in duplicate for each sample, and analyzed by spectrophotometry at 570 nm (Infinite M200 Tecan Multileiter; multi-user laboratory LAMEB/UFSC). The results were calculated as optical density × 1000 mg⁻¹ tissue and expressed as percentage of the controls.

2.6. Measurement of respiratory chain complex activities

The respiratory chain enzyme activity of complexes I, II and IV was measured in homogenates of brain samples, after fGLY exposure for 7 d. A pool of four brains was used per sample (n = 6 samples per group), which were rapidly dissected and homogenized in 10 vol of buffer (300.0 mM sucrose, 5.0 mM MOPS, 1.0 mM EGTA, and 0.1% bovine serum albumin, pH 7.4). Subsequently, the homogenates were centrifuged at 3000 × g for 10 min at 4 °C and the supernatant was frozen at -80 °C. An aliquot of the supernatant was used to determine the protein concentration (Lowry et al., 1951). The activity of complex I was measured by the reduction of the ferricyanide dependent of β-nicotinamide adenine reduced dinucleotide (NADH) at a wavelength of 420 nm, as described by Cassina and Radi (1996). The activity of complex II was measured by the oxidation of 2,6-dinitrophenolindofenol (DCIP) read at 600 nm, according to Fischer et al. (1985). Complex IV was assayed by measuring cytochrome c oxidation at 550 nm (Rustin et al., 1994). The enzymatic kinetics were evaluated at 37 °C for 5 min in a spectrophotometer (Multileitora Infinite M200 Tecan; LAMEB/UFSC), and expressed as nmol/min/mg of protein (Latini et al., 2005).

2.7. Levels of gene transcripts assessed by RT-qPCR

The levels of genes encoding proteins of the mitochondrial respiratory chain, *ndufa6* (complex I), *sdhc* (complex II) and *cox1* (complex IV) were analyzed in the zebrafish brain after fGLY exposure for 7 d. The extraction of total RNA and cDNA synthesis were conducted according to Jaramillo et al. (2017). For the extraction of total RNA, a pool of two brains per sample (n = 3 samples per group) was used. For cDNA synthesis, 1.0 μg of total RNA from samples and the GoScript™ Reverse Transcriptase kit (Promega) were used according to the manufacturer's instructions. Specific primers were designed using Oligo Explorer v1.5 (Gene Link), and primer-dimers and hairpins were evaluated by the Oligo Analyzer v1.5 (Gene Link). In addition, primers described for the *rpl8* reference gene were used (Uren Webster et al., 2014) (Table 1). All reactions were performed in 96-well plates using the 7900^{HT} Fast Real-Time PCR System (Applied Biosystems). The qPCR reaction (10.0 μL) was performed in duplicate by sample using the GoTaq® qPCR Master Mix (Promega), which consisted of 5.0 μL GoTaq® Master Mix 2×, 0.3 μM of each primer, 0.1 μL CXR Reference Dye, 1.0 μL of diluted cDNA (1/10) and nuclease-free water. The qPCR cycling conditions were 1× (95 °C for 10 min), 40× (95 °C for 15 s and 62 °C for 1 min) and dissociation curve (95 °C for 15 s, 60 °C for 15 s, 95 °C for 15 s). The negative control of each gene was included in each qPCR plate. Transcript levels for genes in the brain were calculated using the 2^{-ΔΔC_q} method (Livak and Schmittgen, 2001) using *rpl8* as the reference gene (Jaramillo et al., 2017).

2.8. Evaluation of mitochondrial transmembrane potential

Changes in the mitochondrial membrane potential were evaluated using the cationic dye JC-1 (5',6,6'-tetrachloro-1,1',3,3'-tetraethylbenzimidazolylcarbocyanine iodide, Thermo Fisher Scientific). After 7 d of fGLY exposure, the brains (n = 12 samples per group) were dissected and homogenized in 250.0 μL of balanced salt solution buffer (BSS) (136.0 mM NaCl, 5.4 mM KCl, 1.4 mM MgCl₂, 1.0 mM NaH₂PO₄, 1.2 mM CaCl₂, 10.0 mM HEPES (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid), 9.0 mM glucose, pH 7.4). After 3 min of homogenization, 100.0 μL the supernatant containing cells was collected and treated with 2.0 μM JC-1 at 27 °C for 20 min in the dark. Then, the cells were centrifuged at 1000 × g, for 2 min, the pellets were suspended in 220.0 μL of BSS

Table 1
Genes and primer sequences for RT-qPCR analysis.

Gene		Primer sequences (5'–3')	Amplicon (bp)	Amplification efficiency
Symbol	Name			
<i>ndufa6</i>	NADH dehydrogenase (ubiquinone) 1 alpha subcomplex 6	AGGTGTGGAAGCAGAGGA GAATTTGGAGAGGAAGTCAGC	83	1.94
<i>sdhc</i>	succinate dehydrogenase complex, subunit C, integral membrane	AGAGGAACAGGCATAGCAC CTAACACCAACGCAGCAAG	64	1.94
<i>cox1</i>	cytochrome c oxidase I, mitochondrial	TGAACCTTAGCCAACCGAGAG GGGTGGAAGAAATCCAGAG	205	1.94
<i>rpl8</i>	ribosomal protein L8	CCGAGACCAAGAAATCCAGAG CCAGCAACAACCAACAAC	91	1.93

buffer, and 100.0 μL of each sample was added to a 96-well plate, in duplicate, and analyzed using a fluorescence spectrophotometer (Spectramax Paradigm; LAMBE/UFSC) at 488 nm excitation and 529 nm and 590 nm emission (green and red, respectively). The JC-1 red/green fluorescence ratio was calculated and expressed as a percentage of the control. For the positive control of the reaction, 50.0 μM CCCP (carbonyl cyanide 3-chlorophenylhydrazone), a mitochondrial membrane potential disrupter, was added to the cells before treatment with JC-1 and incubated at 27 °C for 5 min.

2.9. Measurement of reactive species

Reactive species (RS) formation was detected using 2',7'-dichlorofluorescein diacetate (DCFDA, Sigma-Aldrich). After 7 d of fGLY exposure, the brains ($n = 12$ samples per group) were dissected and homogenized in BSS according to section 2.8. The cells were collected and treated with 10.0 μM DCFDA (final concentration) at 27 °C for 30 min in the dark. Then, the cells were centrifuged at 1000 $\times g$ for 2 min, the pellets were suspended in 200.0 μL of BSS buffer, and 100.0 μL of each sample was added to a 96-well plate, in duplicate. The formation of the oxidized fluorescent derivative (DCF) was monitored at 488 nm excitation and 525 nm emission using a fluorescence spectrophotometer (Spectramax Paradigm; LAMBE/UFSC). DCF fluorescence was corrected according to the protein content (Lowry et al., 1951) and expressed as percentage of control.

2.10. Behavioral evaluation

To perform the behavioral analysis of zebrafish, a Ctrax Caltech Multiple Fly Tracker[®] was used to evaluate exploratory activity and aggregation behavior (adapted according to Barry, 2012). After 7 d of fGLY exposure, animals ($n = 18$ animals per group) were transferred to a 20 L aquarium and acclimated for 10 min prior to video recording. Spontaneous free swimming was recorded for 10 min. The relative X and Y positions of each fish in each frame (a total of 500 frames) was analyzed using the Behavioral MicroArray Toolbox.

2.11. Statistical analysis

The results are presented as mean $\pm$ standard error of mean (SEM). Data were analyzed using one-way ANOVA, followed by Dunnett's *post hoc* test. Differences among groups were considered significant when $P < 0.05$.

3. Results

3.1. Acute toxicity of GBH in zebrafish

The first analysis shown here is the acute exposure of zebrafish

D. rerio to fGLY. The percentage survival of fish up to 96 h showed that the tested concentrations $<30.0 \text{ mg L}^{-1}$ fGLY (0.18 mM) did not cause fish death. The survival curve showed that the fish exposed to the concentrations of 40.0 and 50.0 mg L^{-1} fGLY (0.24 and 0.30 mM, respectively) presented 87.5% and 75.0% survival, respectively. At the concentrations of 60.0 and 70.0 mg L^{-1} fGLY (0.36 and 0.41 mM, respectively), survival was 25.0% and 12.5%, respectively. The 100% mortality of fish was observed in two situations, i.e. at 24 h at the concentration of 100.0 mg L^{-1} fGLY (0.59 mM) and at 72 h with 80.0 mg L^{-1} fGLY (0.47 mM). The log-rank test revealed that the survival curves were statistically different ($\chi^2 = 177.1$; $df = 1$; $p < 0.0001$). Additionally, the calculated $\text{LC}_{50,96 \text{ h}}$ for zebrafish was $53.75 (\pm 3.30) \text{ mg L}^{-1}$ fGLY [$0.32 (\pm 0.02) \text{ mM}$] with a 95% confidence interval between 50.4 and 56.9 mg L^{-1} fGLY (Fig. 1).

3.2. In vitro and in vivo effects of GLY and fGLY on brain cell viability

In vitro and *in vivo* assays were performed in order to better recognize the effects of GLY and fGLY on cell viability. After 3 h of *in vitro* incubation with fGLY, a significant decrease in brain cell viability occurred starting at $1.6 \times 10^3 \text{ mg L}^{-1}$ fGLY (10.0 mM), while incubation with GLY induced a significant reduction starting at $4.0 \times 10^3 \text{ mg L}^{-1}$ (25.0 mM). Concentrations of 0.065 (0.38 μM), 1.0 (6.0 μM), 10.0 (60.0 μM), and 160.0 (1.0 mM) mg L^{-1} of GLY and fGLY did not induce a reduction in cell viability. *In vivo* exposure to 0.065 and 1.0 mg L^{-1} fGLY for 7 d induced a significant reduction of cell viability, when compared to the control group. However, no change in cell viability was observed at 10.0 mg L^{-1} fGLY (Fig. 2).

3.3. Effects of exposure to fGLY on respiratory chain complex activities

After recognizing the effects of fGLY on cell viability, we analyzed its effects on the activity the enzymes NADH dehydrogenase (complex I), succinate dehydrogenase (complex II), and cytochrome C oxidase (complex IV) in the brain after 7 d of exposure. The concentrations of 0.065 and 1.0 mg L^{-1} fGLY induced a significant inhibition in the activity of NADH dehydrogenase and cytochrome C oxidase compared to the control. However, regarding the enzymatic activity of succinate dehydrogenase, no change was observed among the exposed and control groups. Interestingly, at 10.0 mg L^{-1} fGLY, there were no observed alterations in enzyme activity (Fig. 3).

3.4. Effects of fGLY on transcript levels of the genes involved in the mitochondrial respiratory chain

Since effects of fGLY on enzyme activity were observed, the transcript levels of the genes encoding for electron transfer proteins (*ndufa6*, *sdhc*, and *cox1*) were also evaluated in the brain after 7 d of exposure. A significant increase in transcripts of *ndufa6* was

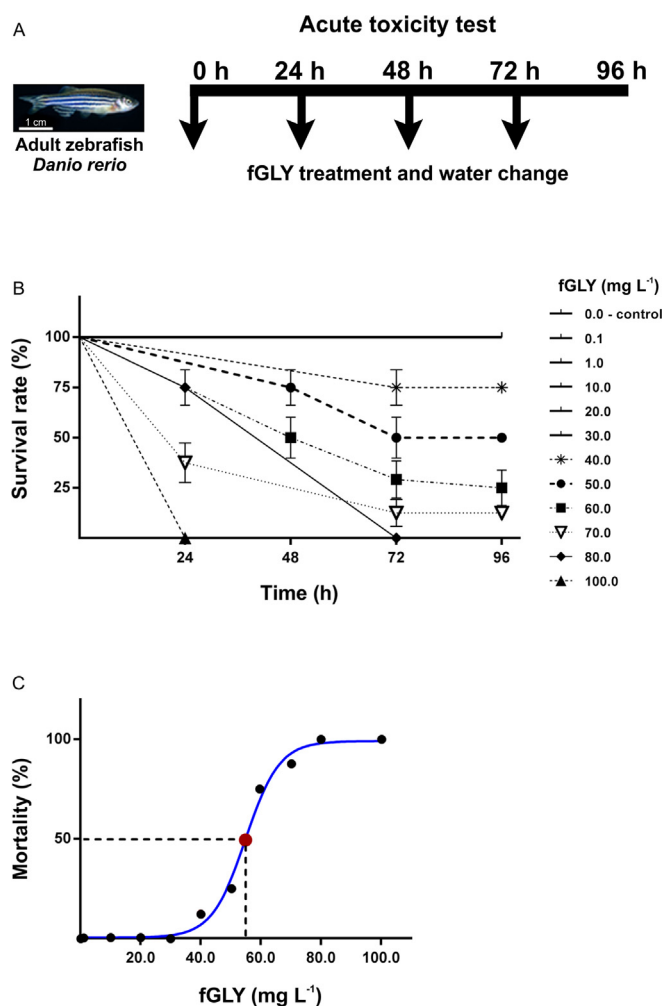


Fig. 1. Acute toxicity test of fGLY using adult zebrafish *Danio rerio*. Experimental design for the acute toxicity test (A). Survival rate of fish registered during the 96 h of acute toxicity test (B). The points on the curve of graph B indicate the mean percentage survival in every 24 h period for each tested concentration. Dose-response curve of fGLY exposure at 96 h (C). The points on the curve of graph C indicate the mean percentage of mortality for each tested concentration. The red point on the edge of the dashed lines indicates the LC_{50,96h}.

observed at all concentrations tested (0.065, 1.0, and 10.0 mg L⁻¹ fGLY) in comparison to the control. The transcripts of *sdhc* showed a significant decrease only at 0.065 mg L⁻¹ fGLY, while the transcripts of *cox1* were significantly decreased at 1.0 and 10.0 mg L⁻¹ fGLY (Fig. 3).

3.5. Effects of fGLY on mitochondrial membrane potential ($\Delta\Psi_m$) and RS formation

The effects of fGLY on mitochondrial membrane potential were evaluated, considering the important aspects of the functionality of this organelle. We observed a significant increase in mitochondrial membrane potential in the zebrafish brain at all tested concentrations (0.065, 1.0, and 10.0 mg L⁻¹ fGLY) in comparison to the control, indicating that fGLY induces hyperpolarization of the mitochondrial membrane (Fig. 4A). Considering the multiple effects of fGLY on mitochondria, we asked whether these effects were associated with changes in the balance of reactive species in brain cells after exposure to the herbicide. Our results show that fGLY induced a significant increase in RS formation at all tested

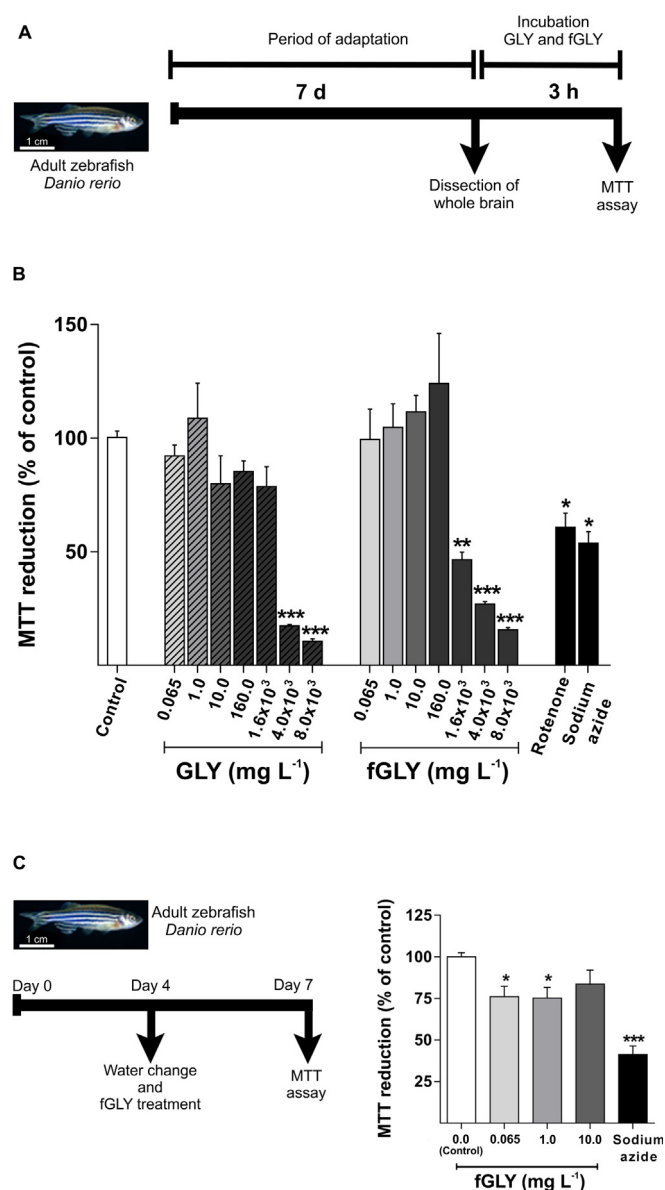


Fig. 2. Effects of GLY and fGLY exposure on cell viability using zebrafish *Danio rerio* brain. Experimental design for *in vitro* assay (A). *In vitro* effects of tested concentrations of GLY and fGLY on cell viability after 3 h of incubation (B). Rotenone (100.0 μ M) and sodium azide (50.0 mM) were used as positive controls for the MTT assay. Experimental design for *in vivo* assay (C). Effects of fGLY on the cell viability after 7 d of exposure (D). Sodium azide (50.0 mM) was used as the positive control. Bars represent the mean $\pm$ SEM. * P < 0.05; ** P < 0.01 and *** P < 0.001, when compared to control.

concentrations (0.065, 1.0, and 10.0 mg L⁻¹ fGLY), which was quantified after 7 d of exposure (Fig. 4B).

3.6. Effects of fGLY on behavior

The final analysis was conducted in order to determine whether fGLY is able to impair the behavior of exposed fish. After 7 d, in contrast to the controls, animals exposed to 0.065 mg L⁻¹ fGLY showed exploratory activity mainly in the top of aquarium and did not show the typical aggregated behavior (Fig. 5). Animals exposed to 1.0 and 10.0 mg L⁻¹ fGLY stayed in the bottom of the aquarium and animals exposed to 10.0 mg L⁻¹ fGLY showed a reduction in exploratory activity.

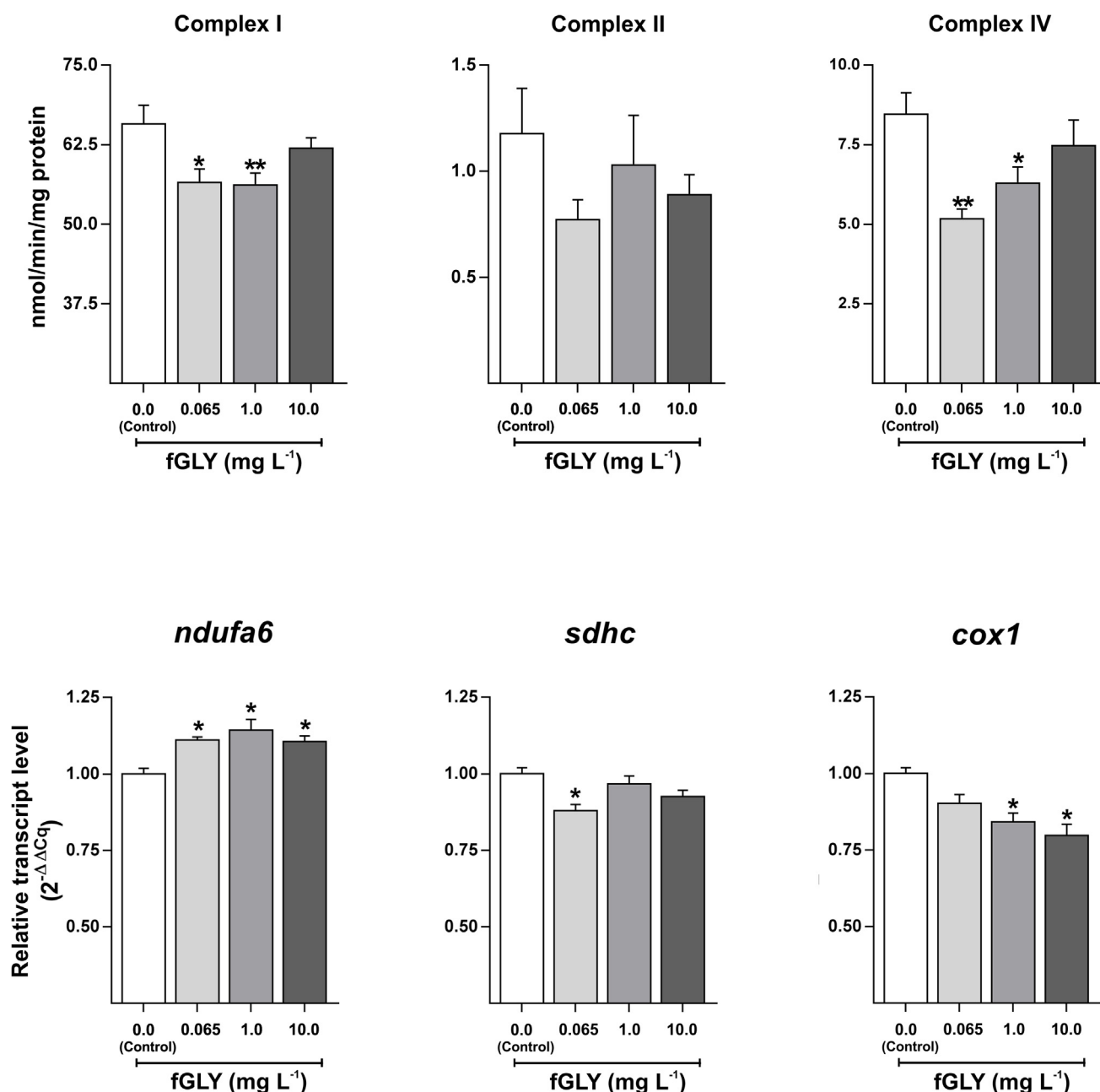


Fig. 3. Respiratory chain changes in cells of zebrafish *Danio rerio* brain after 7 d of fGLY exposure. Activity of complexes I, II and IV of the respiratory chain and transcript levels of *ndufa6*, *sdhc* and *cox1* genes. Data on enzymatic activity represented as the mean $\pm$ SEM. Data on transcript levels were normalized using the *rpl8* reference gene and the results are presented as percentages of the control. * $P < 0.05$; ** $P < 0.01$, when compared to control.

4. Discussion

GLY and GBH are soluble agrochemicals that can affect non-target organisms, such as human workers and aquatic animals, and potentially cause important noxious effects. Unfortunately, these effects are not well-understood and the conflicting information available in the current literature does not contribute to an appropriate interpretation of the real effects of these agrochemicals. Here, we propose an integrative approach, which combines different and complementary assays to demonstrate the effects of fGLY on the brain, using zebrafish as a model organism. The most important contribution of this study was the observation of different levels of mitochondrial dysfunction, ranging from alterations in the enzymatic activity and transcript level of complexes I,

II and IV, hyperpolarization of the mitochondrial membrane, and decreased cell viability evaluated by mitochondrial metabolic activity after 7 d of fGLY exposure.

Of the different tests carried out, the survival test is usually the first approach to evaluate the environmental safety and toxicity of chemical substances (Rand et al., 1995). Here, we provide essential data on the LC_{50,96 h} (53.75 mg L⁻¹ fGLY) for zebrafish (*D. rerio*) under laboratory conditions. LC_{50,96 h} has been obtained for other GBH formulations in fish (not zebrafish), and generally vary according to the fish life stage, body size, water physicochemical parameters, absorption rate, and detoxification mechanisms, as well as differences in the composition of the tested formulations (Rand et al., 1995; Tsui and Chu, 2003). To establish a closer comparison, we choose assays performed in other fish with a similar

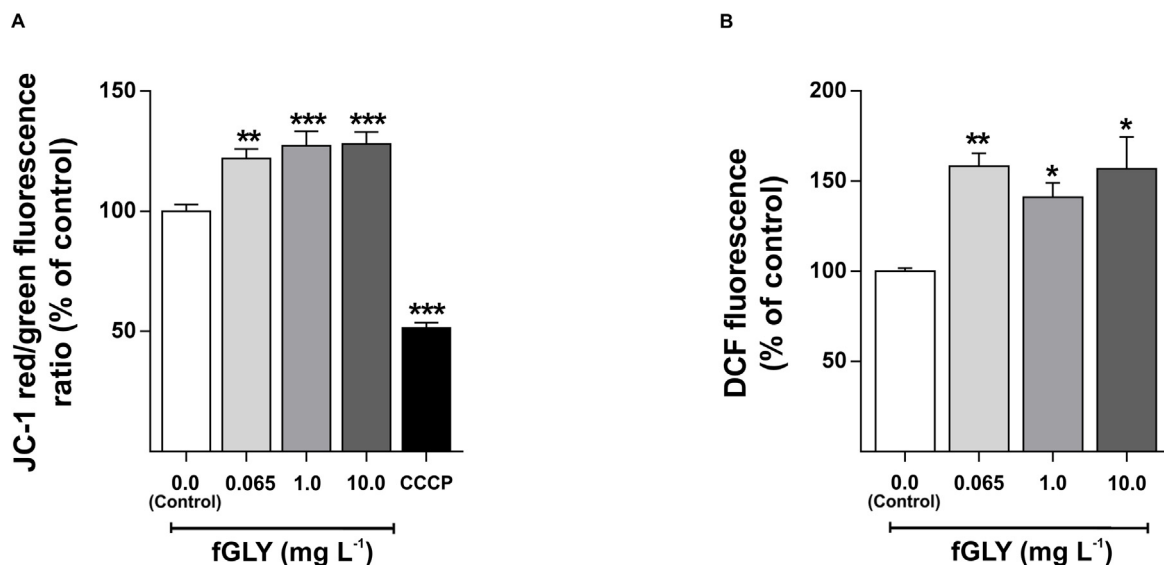


Fig. 4. Effect of fGLY on toxicological classical parameters in the zebrafish *Danio rerio* brain after 7 d of exposure. Mitochondrial membrane potential using CCCP as the positive control (A). ROS formation (B). The results are presented as percentages of the control. Bars represent the mean $\pm$ SEM. * $P < 0.05$; ** $P < 0.01$ and *** $P < 0.001$, when compared to control.

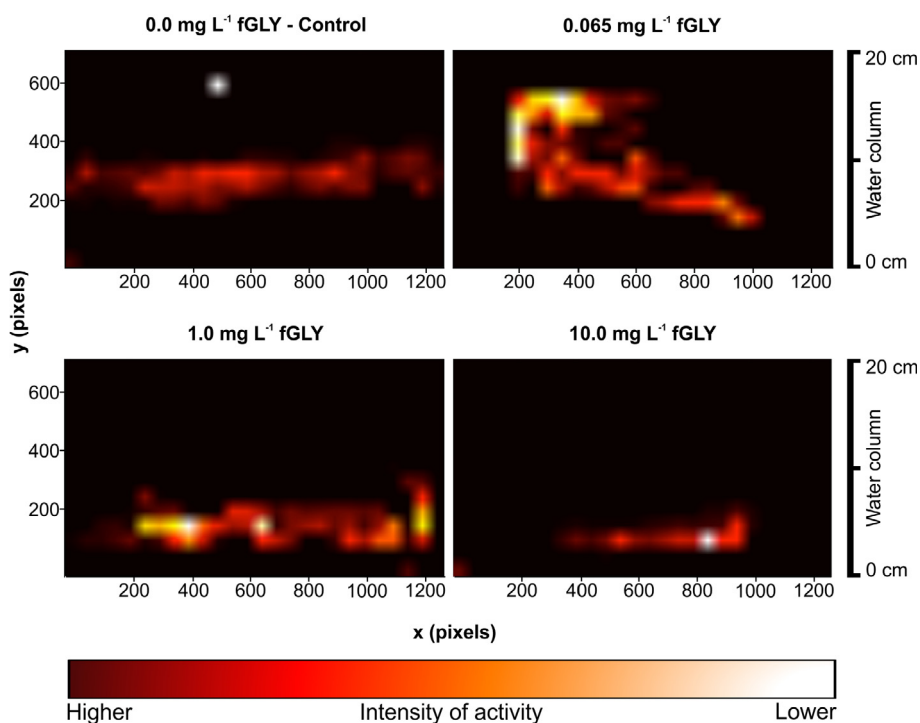


Fig. 5. Behavioral impairments induced by 7 d of fGLY exposure. The exploratory activity was evaluated by the position of the fish in the aquaria (width and height). The aggregation behavior was evaluated by the color intensity of the shadow. Colored shadow represents exploratory activity and aggregation behavior of fish in water column at aquarium. Color bar indicates the intensity of fish activity. Vertical line indicates the height of the water column.

body length as *D. rerio* (3.3 cm, tested here), i.e. the guppy *Poecilia reticulata* (3.21 cm) with an LC_{50,96 h} of 3.6 mg L⁻¹ fGLY (Rocha et al., 2015) and the rainbow trout *Salmo gairdneri* (4.9 cm) with an LC_{50,96 h} of 54.8 mg L⁻¹ fGLY and 52.0 mg L⁻¹ fGLY under laboratory and field conditions, respectively (Hildebrand et al., 1982). Folmar et al. (1979) also proposed the LC_{50,96 h} for four aquatic invertebrates, using GLY, GBH and surfactants, showing a range of toxicity tolerance, which can be considered for GBH and GLY risk assessment evaluations. Therefore, due the lack literature on the

LC₅₀ for GBH in zebrafish and variations depending on the calculation of LC₅₀, we emphasize the importance of providing these essential data for the GBH used in the present study.

In this study, taking into account that the evaluation of sublethal concentrations is important to establish the toxic potential of environmentally relevant compounds, we investigated the effects of *in vitro* and *in vivo* exposure of ca. 0.12%, 2.0% and 20.0% of the LC_{50,96 h} of fGLY in the zebrafish brain. We found that *in vitro* fGLY treatment reduced the viability of brain cells at lower

concentrations that those necessary to induce the loss of viability with GLY. This result agrees with other findings showing that some formulations are more toxic than pure GLY (Tsui and Chu, 2003; Benachour and Séralini, 2009). Folmar et al. (1979) suggested that the surfactants present in GBH formulations are responsible for the greater toxicity in comparison to pure GLY. According to Howe et al. (2004), the differences among formulations depend on the amount of surfactant, but the surfactant information for some of these compounds can be difficult to obtain. The most common surfactant contained in GBH is polyethyloxylated tallow amine (POEA), which has been described as more toxic than the active ingredient itself (Folmar et al., 1979; Mesnage et al., 2013).

The lowest concentrations (0.065 and 1.0 mg L⁻¹ fGLY) used in the *in vitro* assay for 3 h did not induce changes in cell viability. However, these same concentrations *in vivo*, after 7 d of exposure, induced a decrease in cell viability. According to these data, we suggest that the decrease in brain cell viability *in vivo* was induced as a chronic or sub-chronic systemic response. The longer exposure time to fGLY may have induced greater toxicity in the brain, as well as allowing the pro-viability response at the highest concentration tested (10.0 mg L⁻¹ fGLY). According to Peixoto (2005), higher concentrations or a longer exposure period should probably be used in order to obtain a better correlation between *in vivo* and *in vitro* results. This author also affirms that *in vitro* assays that considerer viable mitochondria cannot be longer than a few hours; in this case, only acute effects can be studied.

Mitochondrial dysfunction is a hallmark of neurodegeneration (Nunnari and Suomalainen, 2012; Rugarli and Langer, 2012), and some pesticides exert their neurotoxic effects on proteins that constitute the electron transport chain (ETC), modulating their activity (Zhang et al., 2003; Bjørling-Poulsen et al., 2008). The reduced metabolism indicated in our MTT assays suggests that fGLY changed mitochondrial activity in the zebrafish brain *in vivo*, mainly in groups exposed to lower concentrations. Therefore, for the first time to our knowledge, we have demonstrated the effects of fGLY on the activity of complexes I and IV of the ETC, as well as on transcript levels for the key enzymes of these complexes in brain of zebrafish, showing the effects this herbicide on mitochondria. The partial inhibition of mitochondrial complexes was observed in isolated rat liver mitochondria exposed to GBH (15.0 mM), higher than used in our *in vivo* study, and pure GLY (50.0 mM) induced significant inhibition of mitochondrial bioenergetics (Peixoto, 2005). Olorunsogo et al. (1979) and Olorunsogo (1990) suggest that pure GLY induces an uncoupling of mitochondrial oxidative phosphorylation and an increase in mitochondrial membrane permeability in rat liver cells. Similar changes in gene expression observed in complex I and II were also described by Mesnage et al. (2015) in the liver and kidney of rats following ultra-low dose exposure to GBH (50.0 ng L⁻¹).

The ETC is one of the most important sites of ROS production under physiological conditions (Tan et al., 1998; Li et al., 2003). Inhibition of complex I and IV and mutation of their genes promote an increase in mitochondrial ROS production (Bolter and Chefurka, 1990; Li et al., 2003; Abramov et al., 2010). In this context, considering the changes observed in mitochondrial complexes, we investigated whether fGLY exposure resulted in increased RS production. Our results indicate that changes in mitochondrial complexes are associated with increased RS formation, in accordance with the results of other studies involving GLY and/or fGLY exposure. Changes consistent with the generation of oxidative stress and the widespread induction of compensatory cellular stress response pathways in liver of *Salmo trutta* were similar with both GLY and GBH treatment (Webster and Santos, 2015). Moreover, GBH was also reported as an inducer of oxidative stress in hippocampus (Cattani et al., 2014, 2017) and liver (El-Shenawy, 2009) of rat, as

well as in worms *C. elegans* (Bailey et al., 2018).

CNS function strongly depends on efficient mitochondrial function, because brain tissue has a high energy demand (German et al., 1987). Considerable evidence in the literature support the possibility that mitochondrial dysfunction and oxidative stress play a crucial role in neurodegenerative diseases (Federico et al., 2012). Neuronal health critically depends on the integrity and functionality of mitochondria (Rugarli and Langer, 2012). Inhibition of the mitochondrial ETC reduces proton efflux across the mitochondrial inner membrane and is generally associated with $\Delta\psi_m$ depolarization, but partial complex I inhibitory conditions have been linked to $\Delta\psi_m$ hyperpolarization (Barrientos and Moraes, 1999; Forkink et al., 2014). Therefore, considering the changes observed in mitochondrial complexes associated with increased RS production, here we demonstrate that exposure to fGLY for 7 d also induces $\Delta\psi_m$ hyperpolarization in brain cells. The inhibition of complex I and enhanced $\Delta\psi_m$ are classical inducers of ROS generation in mitochondria (Abramov et al., 2007). Various pathological conditions, including mitochondrial complex I deficiency, have been linked to $\Delta\psi_m$ hyperpolarization (Abramov et al., 2010; Forkink et al., 2014), although the mechanism is not fully understood. Moreover, the $\Delta\psi_m$ hyperpolarization in complex I-inhibited cells results from low activity of complex II, III and IV, combined with reduced forward action of complex V (Forkink et al., 2014). Based on our data, we suggest that fGLY induces mitochondrial dysfunction in brain cells after 7 d exposure, evidenced by the decrease in the activity of complexes I and IV, associated with changes in the transcript levels for the key enzymes of complexes I, II and IV, mainly in the groups exposed to the lowest concentrations (0.065 and 1.0 mg L⁻¹ fGLY), followed by an increase RS production and $\Delta\psi_m$ hyperpolarization.

Based on the results obtained in our study, we also investigated the behavior of zebrafish. Our results show that fGLY exposure for 7 d induced behavioral alterations, evidenced mainly in the group exposed to the lowest concentration tested. These results could be the consequence of mitochondrial dysfunction, RS production, and reduced cell viability in the brain, as shown in this study. Recently, Bridi et al. (2017) also found that GBH alters locomotion and aversive behavior in zebrafish larvae and adults, and leads to memory impairment in adults.

In conclusion, we have demonstrated the neurotoxic effects of the fGLY (mainly 0.065 mg L⁻¹) in the brain of zebrafish (*D. rerio*), with an increase in RS production, changes in the levels of mitochondrial complexes and hyperpolarization of the mitochondrial membrane, which could lead to a decrease in cell viability and finally in behavioral alterations. Mitochondrial dysfunction is recognized as a relevant cellular response that should not be disregarded. Moreover, this study points to mitochondria as an important target of GBH. These results contribute to our knowledge of the effects of GBH on animal brains and encourage further studies at the molecular level to reveal its mechanism of action and to revise the regulations on the maximum limits permissible for this herbicide in water.

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